

2nd TEST ANSWER SHEET

MATRICULATION :

FULL NAME :

GROUP:

EXERCISE 1 (5 POINTS).

Main Program	ALGORITHM Exo1 TYPE bit = 0..1 CONST Tmax = 16 BEGIN VAR pos, lg, x:integer T, TR: array [Tmax] of bit <i>//(1) declare the binary array</i> scan (x) <i>//(2) read the decimal number</i> <i>//(3) print its SVA representation</i> DECtoSVA(T, x, Tmax); printf("its SVA representation is "); BINPRINT(T, Tmax); <i>//(4) print if the binary number is even or odd</i> if (BINPARITY(T, Tmax)) then printf("\nthe binary number is even") else printf("\nthe binary number is odd") <i>//(5) print its double in binary</i> BINDOUBLE(T, TR, Tmax) printf("its double in binary is ") BINPRINT(TR, Tmax) <i>//(6) print the starting index of the longest consecutive sequence of zeros and its length.</i> LGSTSEQ(T, Tmax, &pos, &lg) printf("\nthe longest consecutive sequence of zeros start in, pos, "position and its length is", lg) END
WRITE1D Module	PROCEDURE BINPRINT (T: array [] of bit, N: integer) Var i: integer Begin for i ← 0 to N do print (T[i]); End

DECtoSVA Module	<pre> PROCEDURE DECtoSVA(Var T: array [] of bit, x, N: integer) Var I : integer BEGIN if (x>0) then T[0] $\leftarrow$ 0; else BEGIN T[0] $\leftarrow$ 1 x $\leftarrow$ (-1) * x END for i $\leftarrow$ N-1 to 1 step -1 Do BEGIN T[i] $\leftarrow$ x mod 2; x $\leftarrow$ x div 2; END END END </pre>
BINPARI TY	<pre> FUNCTION BINPARITY(T: array [] of bit, int n): Boolean BEGIN return(T[n-1]=0); END </pre>
BINDOUBLE module	<pre> PROCEDURE BINDOUBLE(T : array [] of bit, Var TR: array [] of bit, n: integer) VAR i : integer BEGIN TR[0] $\leftarrow$ T[0]; //T[0] bit de signe il ne change pas for i $\leftarrow$ 1 to n-2 do TR[i] $\leftarrow$ T[i+1]; TR[n-1] $\leftarrow$ 0; //la multiplication par 2 END </pre>
LGSTSEQ module	<pre> PROCEDURE LGSTSEQ(T :array [] of bit, n: interger, VAR pos, Var lgst: integer) VAR posc,lgc, i : integer BEGIN posc $\leftarrow$ -1; lgc $\leftarrow$ 0; pos $\leftarrow$ -1 ; lgst $\leftarrow$ 0; for i $\leftarrow$ 0 to n-1 do BEGIN if ((T[i] = 0) AND (lgc>0)) then lgc $\leftarrow$ lgc + 1; if ((T[i] = 0) AND (lgc=0)) then BEGIN lgc $\leftarrow$ 1; posc $\leftarrow$ i; END if (T[i] = 1) then BEGIN if (lgc>lgst) {lgst $\leftarrow$ lgc; posc $\leftarrow$ posc;} lgc $\leftarrow$ 0; posc $\leftarrow$ -1; END END END END END </pre>

EXERCISE 2 (5 POINTS).

ALGORITHM MULMIRROR

FUNCTION NBOCC(i, c: integer): integer

FUNCTION CH1A9_DIFF(i: integer): boolean

FUNCTION MIROIR(n: integer): integer

VAR i, a, b, ma, mb: integer

BEGIN

for i $\leftarrow$ 12345 to 98765 do

if (CH1A9_DIFF(i)) then

BEGIN

b $\leftarrow$ i mod 100; ma $\leftarrow$ MIROIR(b);

a $\leftarrow$ i div 100; mb $\leftarrow$ MIROIR(a);

if ((a * b) = (mb * ma))

printf(a, "*" , b , " = " , mb, "*", ma, "=", a*b);

END

END

FUNCTION NBOCC(i, c: integer): integer

VAR cpt: integer

BEGIN

cpt $\leftarrow$ 0

while (i > 0) do

begin

if ((i mod 10) = c) cpt ++;

i $\leftarrow$ i div 10;

end

return cpt;

END

FUNCTION CH1A9_DIFF(i: integer): boolean

Var c: integer; uniq : boolean

BEGIN

uniq $\leftarrow$ (NBOCC(i, 0) = 0)

c $\leftarrow$ 1

while ((c < 9) AND (uniq)) do

BEGIN

if (NBOCC(i, c) > 1) then uniq $\leftarrow$ false;

c++;

END

return uniq;

END

FUNCTION MIROIR(n: integer): integer

Var m : integer

BEGIN

m $\leftarrow$ 0;

while (n > 0) do

Begin

m $\leftarrow$ m * 10 + (n mod 10);

n $\leftarrow$ n div 10;

END

return m;

END